RESEARCH ASSIGNMENT 1

1) A database is a digital filing cabinet, a collection of related information. They are essential in a efficient access and management of information.

Two real-world examples of where databases are used:

- Whatsapp / imessages, where all your chats and contacts are stored.
- Shein, where your orders and customer ID is stored.

2) A data warehouse is where valuable data is stored, e.g sales data, customer data, inventory etc.
A regular database is designed to handle day-to-day operations (storing real time data) while a data warehouse is large and designed for online analytical processing.
A company would use a data warehouse for storing historical data ~~and~~ as well as data analysis and reporting.

3.) A data lake is designed to store large volumes of raw data in its orignal form. It is to accommodate all types of data from any source. A data lake stores both relational and non-relational data from a variety of sources whereas a data warehouse stores data that has been processed and is structured. A business would prefer a data lake for machine learning and AI initiatives, raw data storage for model training.

4.) A data mart is a data storage system that contains information relevant to a particular department. A datamart can be sourced from a data warehouse ensuring data consistency.
A data mart is typically used by Sales, HR, marketing to gain insight and decisions based on department specific data.

5.) structured data : refers to data stored in a table with rows and columns [A consistent format) e.g. a spreadsheet

Semi structured data : refers to data that doesn't live in neat tables but it still has keys or tags that tell you ~~which~~ what each piece of data means.
Unstructured data : refers to data that is not stored in a specific manner (inconsistent format) e.g. a social media post, podcast.

6) Data modeling is the process of creating a visual representation of how data will be stored.

Two types of data models:
- Physical data models: Is how data is stored
Tell the database exactly how to build it.

Conceptual data models: Is what data exists
What data do we care about.

7) Data mining is the process of discovering patterns in large datasets. In a business data mining provides a way to turn raw information into knowledge that supports better decisions. Examples of data mining:

- fraud detection
- Personalised recommendations.

8) A DBMS is a system that allows us to interphase our data. We use it to interact with our data.

Four popular DBMS platforms:

- PostgreSQL: Tech companies, data analysts
- My SQL: Web apps, Word Press, startups
- Oracle DB:
- MongoDB: for modern web apps, APIs

9) ETL extract data from sources, transform it then load the cleaned result into the target warehouse.

Source systems

- Where data is created

Transformation

- for cleaning, joining and modeling data

Data warehouse

- where the cleaned, structured analytics are stored.

10) - CSV: A simple format where each row is a record and columns are separated by commas

- SQL: represents data stored in a relational database.
- Parquet: optimised for big data analytics, stores data column by column for compression.

I would use CSV for simplicity, SQL for structured data and Parquet for big data processing.

SECTION B: ARTIFICAL INTELLIGENCE

11) Artificial intelligence is the ability of a computer to perform tasks commonly associated to human intelligence. Narrow AI is designed for specific tasks and is limited to its programmed capabilities, e.g. phone apps where as general AI is designed to replicate human intelligence such as learning and reasoning. to new situations without direct reprogramming.

12) Machine learning is a type of AI that focuses on building algorithms capable of learning patterns from data and making predictions / decisions without ~~being~~ explicit, hard-coded instructions.

- Supervised learning is a machine learning techinique that uses labeled data sets to train AI models to identify underlying patterns and relationships.
- Reinforcement learning is a type of machine learning process in which agents learn to make decisions by interacting with their enviroment.
- Unsupervised learning works with ~~unla~~ unlabled data to find hidden patterns or correlations.

13) Deep learning is a type of ML that uses multi-layered artifical neural networks to learn complex patterns from large datasets. Deep learning models power most generative AI.

Neural networks are computational models inspired by the human brain that learn patterns from data without direct programming. How do they work

14.) Natural language Processing is how we teach computers to understand human language in both text and speech formats.

Three real-world ~~a~~pplications:

- Siri, for conversational interfaces
- Google translate, for cross-language communication
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15.) Large language model is

16.) Generative AI is AI that can create new stuff in response to a users request

Three tools that use generative AI:

ChatGPT

Creates new emails, essays and summaries that didn't exist before

~~Micro~~ Microsoft copilot

Creates slides, formulas, charts, reports from a text prompt

17) A training dataset is a big collection of examples we show to an AI so it can learn patterns and get good at a task.

Biased data will make biased decisions and incorrect data will make wrong predictions.

18) Computer vision is where computers are trained to look at photos, videos or documents to figure out what is in them.
Machines break image into pixels then measure pixels and math turns pixels into meaning.

Real-world use cases of computer vision:

X-ray / MRIs

Computer vision highlights possible tumors for radiologists to double-check.

License plate recognition

Toll gates and speed limit detecters camera reads license plate.

SECTION C APIs, Tools & Developer concepts

19) An API is a set of rules that connects apps together. Its like asking Siri for the weather in your area but since siri doesn't know live weather it will ask the weather API.

20) An API key is a secret /unique code that identifies an application or user when making requests to an API. Its purpose is to offer a level of security.

21) REST stands for Representational state transfer.
A REST API is way for apps to communicate over ~~usin~~ the internet using web URLs
GET: retrieves data from the server
PUT: updates or replaces an existing resource
DELETE: Deletes a resource
PATCH: Briefly updates a resource. Modifies specific fields.

22) JSON (Javascript Object Notation) is an open standard file format for storing and transporting data. It is commonly used because of it's simple text that's readable by humans and programming languages.
JSON objects are written inside curly braces.

23.) Prompt Engineering is designing your questions to AI so it understands your goal and context. It is an important skill because better prompts give better answers and less time is wasted.

Bad prompt:

"write email"

~~Goo~~ Improved prompt:

"Write an email ~~of~~, enquiring about an application thats been waitlisted in a formal manner."

24.) A MAP function takes a list, applies 1 rule to each item and gives you back a new list with results

In programming and data processing MAP takes a rule, applies it to every row automatically and returns a new dataset. It transforms / cleans data and extract fields.

25.) Cloud computing is using other softwares and storage over the internet insteading of owning your hardware.

- IaaS (infrastructure as a service)

Provides on demand servers, storage and networking without the need for physical hardware.

- Paas (Platform as a service)

offers tools and services for application development such as databases, middleware and develop

- SaaS (Software as service)

You rent finished software in your browser. You just manage your data and login details, eg. Netflix and Gmail. You are user not a devloper.

26.) Version Control system is a software that saves every version of your files, has complete visibility to the code history and lets you undo change anytime. Git is a VCS that takes snapshots of your files every time you change them so you can track history. It is used by developers and data analysts ~~it tr~~ because it tracks every change, enables teamwork and makes work reproducible.

- Commit
- A snapshot of all your files at one moment in time, with a note about what change.

Branch
- A copy of your project where you can experiment without breaking the main version.

27.) Jupyter Notebook is a web application that allows you to create and share computational documents with code, text and equations. It is commonly used with Python.
It is popular among Data analysts and Data Scientists because data work is exploration, visualisation and explanation. Jupyter lets you do all three in one document without restarting everything.

28.) Python is a easy to learn language that lets you work quickly and integrate systems more effectively.
Python is most popular for data analytics and AI because it's easy to read, works perfectly with Jupyter and has the biggest community. You solve problems faster.

Four python libraries:

- Pandas
  Handles tables. Load Excel, clean, filter, group, merge. data.
- Numpy
  Fast math on arrays or numbers. Math calculator for big data.
- matplotlib / seaborn
  Turn data into graphs
- Scikit - learn
  Machine learning without PHD math. for fraud detection, clustering and predictions.

Section D: Data Ethics, Governance & Careers

29.) Data Privacy is the right of individuals to control how their personal information is collected, stored, shared and used. It is important for business ethics, personal rights and legal compliance.

GDPR protects an individual's personal data. It imposes strict obligations on organisations that process such data. It harmonises data protection across the European Union.

30.) Data ethics refers to the principles and guidelines that govern the collection, production and use of data.

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31) Data governance is the practice of managing an organisation's data to ensure its availability, quality and integrity.

Key components of a good data governance framework:

Data standards, Policies, and Processes
- Set guidelines for data formats, data models, metadata, naming conventions.

Auditing Procedures
- Auditing and record keeping procedures to maintain transparency and explainability.

32.) Bias in AI occurs when systems treat people unfairly due to factors like race, gender or age.

33.)

	roles	skills	Tools
Data analyst	Take clean data to find insights and makes reports	Curious and detail orientated	Excel, SQL, Python, Power BI
Data scientist	Uses stats to predict future and find hidden patterns.	Problem solver	Python, stats, Jupyter
Data engineer	Builds and maintains pipes that move / store data	Builder and reliability obsessed.	AWS, Python, SQL, databases
AI engineer	Take ML models and puts them into real apps / websites	ship things. Care about speed, user, cost	APIs, Python

3a) Business intelligence is using data and tools to answer "What happened in my business?" so managers can make smarter decisions.

Two popular BI tools :

Power BI

- it connects to your data, cleans it and shows it as dashboards making it easier for managers to use facts

Tableau

- Is built for speed and deep exploration. It lets analysts explore and find insights themselves.

Section E : Personal reflection

35) I found the Bias in AI concept interesting / surprising. I've been ignorant, I never would have imagined there is bias in AI.

36) I found Large language model concept most difficult to understand. I used geeks for geeks to try and understand but I still find it difficult to understand hence I couldn't answer it, also the concept of an API key.

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